

Research papers

Incorporating sediment connectivity index into MUSLE model to explore soil erosion and sediment yield relationships at event scale

R. Hao^a, X. Huang^b, Z.W. Cai^a, H.B. Xiao^a, J. Wang^a, Z.H. Shi^{a,*}

^a College of Resources and Environment, Huazhong Agricultural University, Wuhan 430070, China

^b College of Agricultural Science and Engineering, Hohai University, Nanjing 210098, China

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ABSTRACT

Sediment connectivity quantifies the intermediate processes between initial erosion and the corresponding sediment yield. However, fully capturing the variations in sediment connectivity at the event scale and revealing their impact on watershed sediment sources remains challenging. Herein, we established an event-based model to assess sediment connectivity by integrating runoff factors. The spatiotemporal variability of sediment connectivity was evaluated for 58 rainfall events in three heterogeneous watersheds. The degree of sediment connectivity served as an indicator to quantify the proportion of potential sediment transport from units to the watershed outlet, with 1 indicating that the eroded sediment can be fully connected and 0 indicating that it cannot be transported. Moreover, the controls of sediment connectivity on watershed sediment sources and yield were determined. Our results indicated that the watersheds with poor vegetation and dense gullies were characterized by high sediment connectivity during all the rainfall events. The connectivity degree was below 0.5 in approximately 80 % of the watershed area, suggesting that most of the eroded sediment was deposited. Rainfall amount and duration dominated the degree of connectivity of the distal hillslopes, while rainfall intensity exerted a primary control on the transport of sediments from riverbanks to the outlet. In addition, the increase of mismatched area between sediment connectivity and erosion resulted in the decrease of sediment yield. The interaction between sediment connectivity and erosion effectively explained the spatial patterns of sediment sources ($p < 0.05$). Our work confirmed that the coupled erosion model and sediment connectivity predicted the sediment yield accurately. The predictions for different watersheds portrayed that the Nash-Sutcliffe efficiency and Willmott's agreement index were greater than 0.70 and 0.89, respectively. These findings revealed the role of sediment connectivity and are the critical basis to identify watershed-specific sediment management practices.

1. Introduction

In the context of frequent extreme rainfall events associated with climate change, sediment transport at short intervals imposes a greater risk of flooding and land degradation (Dunkerley, 2019). Approximately 10 high-intensity rainfall events could contribute to more than half of the total annual sediment yield, according to statistics from 1314 watersheds (Gonzalez-Hidalgo et al., 2010; Li et al., 2020). Moreover, activated sediment sources that trigger high sediment yields exhibit significant variations in response to different magnitudes of rainfall within watersheds (Xu et al., 2021). Therefore, it is indispensable to identify which areas connect and transport sediment to the outlet during rainfall events. Sediment connectivity is an effective way to reveal the probability and pathways of sediment transfer from source to an outlet

(Bracken et al., 2015). Understanding sediment connectivity variations in different rainfall regimes will provide a scientific basis for meticulous watershed management that could be customized to control soil loss in specific periods and regions.

Sediment connectivity reflects the ability of sediments to be transported through heterogeneous landscapes until they reach a specific feature of the landscape (e.g., stream, outlet, dam, lake) and the comprehensive role of environmental factors on transport processes (Keesstra et al., 2018). Sediment connectivity includes both structural and functional connectivity: structural connectivity is a potential physical connection between patches of the watershed system (Najafi et al., 2021), and functional connectivity highlights the actual consequences of mechanical processes involved in erosion, transport, and deposition (Pearson et al., 2020). A previous study demonstrated a

* Corresponding author.

E-mail address: shizhihua70@gmail.com (Z.H. Shi).

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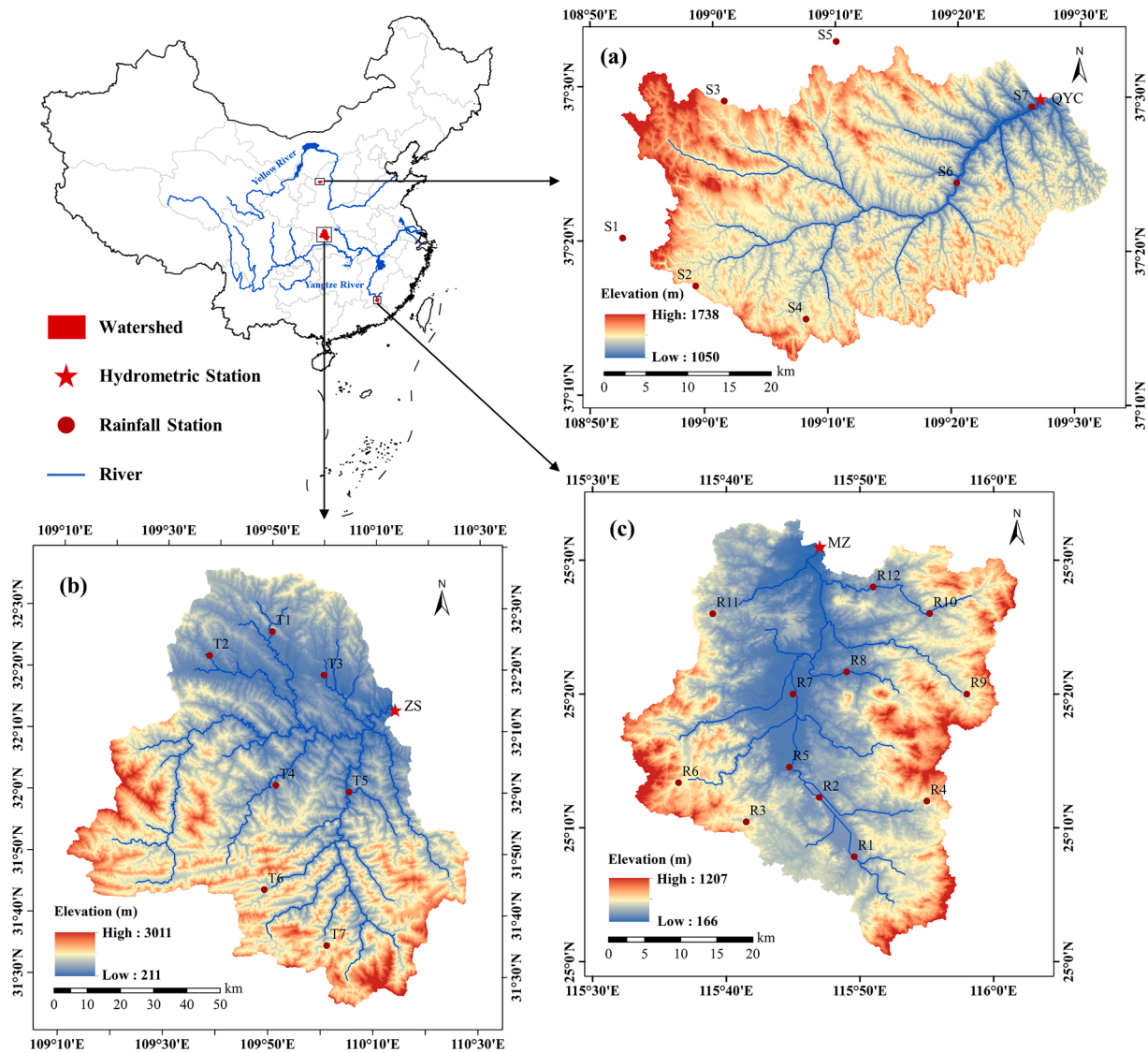


Fig. 1. Locations and digital elevation model (DEM) results of the three watersheds: (a) Dali (DL); (b) Duhe (DH); (c) Xiangshui (XS) watersheds. The main water systems, rainfall stations, and hydrological stations in three watersheds have been marked.

significant positive correlation between sediment yield and sediment connectivity based on the observations of 11 sub-watersheds (Ara-bkhedri et al., 2021). Regions with high sediment connectivity have a high probability of eroding sediment loads reaching the watershed outlet because of low barrier levels (Masselink et al., 2016). Namely, low sediment connectivity decreases sediment accessibility to the outlet, since the cascade relationship between the slope and channel, or bank and channel, is broken (Fryirs, 2013). Furthermore, compared with the traditional lumped sediment delivery ratio (SDR), sediment connectivity emphasizes the spatial information of sediment transport from the source to sink at the fundamental unit scale (Jamshidi et al., 2014). Thus, it attracts huge attention to improve the estimation accuracy of sediment yields, by quantifying and incorporating connectivity into existing erosion models (Michalek et al., 2021). Currently, various methods are available to illustrate sediment connectivity, including field observations, tracer techniques, graph theory, and index methods (Heckmann et al., 2018). The index of connectivity (IC) has been widely validated for multiple scenarios. Borselli et al. (2008) developed IC in regions with vegetation cover or crops. Then, Cavalli et al. (2013) modified it using the roughness index (RI) and demonstrated the capability of topography to control sediment connectivity in small watersheds. Open-source software on sediment connectivity was produced

(Crema and Cavalli, 2018). In addition, the results of the first attempt at soil redistribution rates estimated from coupled IC and soil erosion maps confirmed correlation with field data under six cumulative overflow pathway scenarios (López-Vicente et al., 2015). Some scholars linked IC and SDR, which were available to correct the erosion models used for large watersheds and obtain sediment yield (Vigiak et al., 2012). For example, the IC combined with the revised universal soil loss equation (RUSLE) can be used to determine the spatiotemporal variations in sediment yields caused by land use and dam construction (Zhao et al., 2020). Woznicki et al. (2020) pictured a sediment yield map for the United States of America that linked erosion and downstream transport using the InVEST sediment retention model and IC. These advances utilized sediment connectivity as linkage to explore the nonlinear relationship between erosion and sediment transport and yield.

Modeling sediment connectivity should adequately consider the influencing factors and their weights to improve applicability (Heckmann et al., 2018; Hooke and Souza, 2021). Sediment connectivity is influenced by the internal structure of the watershed, as well as external rainfall conditions (Wainwright et al., 2011). Soil properties and topography affect runoff generation and sediment deposition locations, thereby controlling the spatial patterns of sediment connectivity pathways (Kalantari et al., 2019; Ran et al., 2019). Meanwhile, the

Table 1
Sources and attributes of required datasets.

Data type	Data name	Period	Resolution	Source
Spatial data	DEM	2009	30 m	Aster GDEM V3 Geospatial Data Cloud (https://www.gscloud.cn)
	Soil	2009	Vector data to 30 m	Second National Soil Survey (https://vdb3.soi.lcsdb.cn)
	Land-use	2000–2018	30 m	Resource and Environment Data Cloud Platform (https://www.resdc.cn)
	NDVI	2000–2018	30 m, Monthly	Landsat 7ETM+/8OLI (https://earthexplorer.usgs.gov)
Rain data	Precipitation	2015–2017 (DL) 2000–2010 (DH) 2008–2018 (XS)	Hourly	Annual Hydrological Report of the People's Republic of China, S1-S7, T1-T7, R1-R12 rainfall stations
Hydrological data	Runoff, Suspended sediment content	2015–2017 (DL) 2000–2010 (DH) 2008–2018 (XS)	Hourly	Annual Hydrological Report of the People's Republic of China, QYC, ZS, MZ hydrographic stations

differences in vegetation coverage and land-use patterns cause variations in erosion occurrence and sediment interception capacity (Wu et al., 2020). Watershed areas also affect sediment connectivity and yield owing to different resistance times and transport distances (McGuire and McDonnell, 2010). The reasonability of input data needs to be determined based on the study scales and targets (López-Vicente and Álvarez, 2018). Additionally, rainfall characteristics, such as the rainfall amount, intensity, and duration, play an important role in sediment yield and transport processes (Badoux et al., 2012; Xu et al., 2021). Generally, among the different factors affecting sediment connectivity, soil and topography are relatively stable in the short term, while land cover varies on the seasonal or monthly scale. However, rainfall is unpredictable and drives the extension, contraction, and reconstruction of drainage networks, which complicates the variations in watershed sediment connectivity (Sherriff et al., 2016). Furthermore, most past studies focused on the interannual trends in connectivity patterns, ignoring the distinct variations in sediment connectivity in different rainfall regimes (Hooke and Souza, 2021). Although some studies developed connectivity probability to predict sediment fluxes at the event scale, they cannot compare the degree of sediment connectivity in the connected areas of a watershed (Chartin et al., 2017; Mahoney et al., 2021). Notably, the construction and application of the sediment connectivity framework based on observations of rainfall events are understudied.

Hence, the aim of our study was to propose an updated assessment framework of sediment connectivity, while incorporating erosion models, to predict sediment yield and source distribution at different rainfall event scales. The Modified Universal Soil Loss Equation (MUSLE) model was applied to calculate the amount of eroded sediment at the source (Sadeghi et al., 2014). Then, sediment connectivity served

Table 2
Descriptive statistics of rainfall events in the three watersheds.

Watershed		Rainfall amount (mm)	Rain intensity (mm/h)
Dali	Min	2.65	1.52
	Max	131.03	2.79
	Mean	41.35	10.13
	SD	33.63	10.13
	CV	1.23	0.74
Duhe	Min	50.24	3.69
	Max	145.26	22.8
	Mean	74.8	8.05
	SD	25.55	4.56
	CV	0.34	0.57
Xiangshui	Min	29.96	7.23
	Max	408.56	36.64
	Mean	113.21	18.55
	SD	74.02	8.19
	CV	0.65	0.44

min: minimum; max: maximum; SD: standard deviation; CV: coefficient of variation.

as a bridge to link erosion and sediment yield. We ultimately implemented this assessment framework in three heterogeneous watersheds. The proposed framework mediated the discrepancy between unit erosion and sediment yield measured at the watershed outlet, thus, providing a better perspective on the sensitivity dynamics of sediment sources to different rainfall regimes.

2. Study areas and data collection

2.1. Study areas

We selected three heterogeneous watersheds from north to south in China: the Dali (DL; 37°12'–37°32' N, 108°53'–109°32' E), Duhe (DH; 31°30'–32°37' N, 109°11'–110°25' E), and Xiangshui (XS; 25°00'–25°33' N, 115°34'–111°63' E) watersheds. The DL watershed, located in the Loess Plateau gully region, covered an area of 1310 km² (Fig. 1(a)) (Tian et al., 2019). Yellow loamy soil, known to have poor erosion resistance, was the main soil type in the region. Grasslands with low and medium levels accounted for 46 % of the DL watershed area and drylands covered 44 % of the area. The DH watershed, a mountainous watershed, covered an area of 8973 km² (Fig. 1(b)) (Huang et al., 2018). In this region, the soil types were diverse and included yellowish-brown and brown loam and limestone soil. Land use in the DH watershed was primarily 73 % forest land and 16 % agricultural land. The XS watershed consisted of a hilly region, with a drainage area of 1749 km² (Fig. 1(c)) (Li et al., 2018). The soil was mostly composed of quaternary red soil. Notably, about 83 % of the XS watershed constituted forest land. All the three watersheds experienced a monsoon climate and the average annual rainfall in the DL, DH, and XS watersheds was 382, 990, and 1560 mm, respectively.

2.2. Data collection and pre-processing

Detailed information on the sources and attributes of the datasets is presented in Table 1. The Digital Elevation Model (DEM) from ASTER GDEM V3 data products, with a spatial resolution of 30 m, was used to calculate topographic factors, accumulated flows, and flow directions. The soil maps from the second national soil survey data were transformed into 30-m resolution raster layers, according to relevant studies on the effect of data resolution on connectivity estimation (Arabkhedri et al., 2021). The main soil parameters used were soil unit symbol, texture class, available water storage capacity, relative proportion of different grain sizes, and organic matter. The land use of the three watersheds from 2000 to 2018 was extracted from thematic maps of

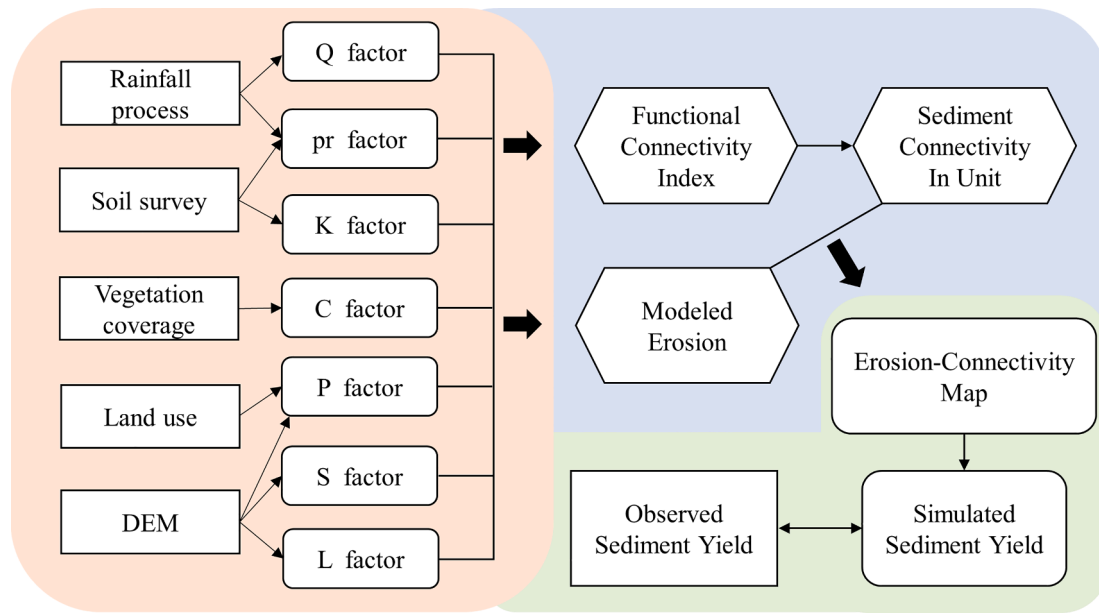


Fig. 2. Schematic representation of the watershed sediment yield model at rainfall event scale.

China's land use. In addition, the normalized difference vegetation index (NDVI) was calculated from Landsat 7ETM+/8OLI images of the corresponding months using Google Earth Engine.

Rainfall and hydrological data during the flooding period were obtained from annual hydrological reports of the People's Republic of China (Table 1). The rainfall data consisted of hourly observations from 26 rainfall stations located in the watersheds (Fig. 1). The rainfall amount and maximum rain intensity were interpolated using the inverse distance weighting method to obtain 30-m precision raster layers in ArcGIS 10.2 (Ruelland et al., 2008). The runoff and suspended sediment content monitored at the QYC, ZS, and MZ hydrological stations were tallied (Fig. 1). Finally, we recorded a total of 10, 18, and 30 events in the DL, DH, and XS watersheds, respectively. The features of these rainfall events are listed in Table 2.

3. Methods

3.1. Model implementation

The erosion and sediment connectivity of each unit were calculated based on the factors considered in the MUSLE model. Sediment connectivity was used as a link to simulate the sediment yield during individual rainfall events in the watersheds. Then, we calibrated the model parameters using the observed sediment yields at the watershed outlet. The overall framework is illustrated in Fig. 2.

3.1.1. Erosion calculation

The MUSLE model was applied to estimate the erosion in individual units, using Eq. (1) given below:

$$E_{unit} = 11.8 \cdot (Q \cdot pr \cdot Area_{unit})^{0.56} \cdot K \cdot C \cdot P \cdot LS \quad (1)$$

where E_{unit} is the potential soil loss for each unit per rainfall (ton); Q is the runoff depth (mm); pr is the peak flow rate (m^3/s); $Area_{unit}$ is the area of each unit (m^2); K is the soil erodibility ($ton\ m^2\ h/m^3\ ton\ cm$); and C , P , and LS are the USLE cover and management, soil erosion control practice, and topographic factors, respectively. K was obtained from soil properties according to the method proposed by Wischmeier and Smith (1978). Furthermore, the obtained NDVI data were used to calculate the C factors for the months corresponding to different rainfall events (Yan et al., 2020). Then, we assigned different P values to different

watersheds according to slope and land use. The LS factor was calculated using the formula proposed by Liu et al. (2002).

where runoff depth (Q) and peak flow rate (pr) were calculated by SCS-CN curve equation (SCS, 2004), as follows Eqs. (2), (3), (4):

$$Q = \frac{(R_{event} - 0.2S)^2}{R_{event} - 0.8S} \quad (2)$$

$$S = 25.4 \left(\frac{100}{CN} - 10 \right) \quad (3)$$

$$pr = \frac{RI}{3600 \cdot R_{event}} \cdot Q \cdot Area_{unit} \quad (4)$$

where R_{event} is the total rainfall amount of an event (mm), S is the potential maximum retention or infiltration (mm), and CN is runoff curve numbers. Q divided by pr is equal to the ratio of the maximum rain intensity (RI , mm/h) to R_{event} , multiplied by the unit conversion coefficient (3600), using the proportionality concept.

3.1.2. Spatial quantification of sediment connectivity

The basic form of the IC (Borselli et al., 2008) can be expressed as follow:

$$IC_k = \log_{10} \left(\frac{D_{up,k}}{D_{dn,k}} \right) = \log_{10} \left(\frac{\bar{W}_k \bar{S}_k \sqrt{A_k}}{\sum_{i=k}^{n_k} \frac{d_i}{W_i S_i}} \right) \quad (5)$$

where the subscript indicates that each unit has its own value. The upslope component ($D_{up,k}$) represents the probability that the sediment generated by the upslope flows to the downstream unit. The downslope component ($D_{dn,k}$) estimates the potential of weighted distance at which the sediments reach the nearest sink along the flow path. W represents the weighing factor (dimensionless), S is the slope (m/m), A is the upslope drainage area (m^2), and d_i is the length of the i th unit along the downslope path.

In this study, the defined W represents the impedance of sediment transport and is derived from the surface characteristics that affect the runoff process and sediment flux in the watershed by combining the factors in the MUSLE. Previous studies indicated that runoff provides energy and carriers for sediment transport (Tetzlaff et al., 2014), thus we

added $11.8 \cdot (Q_i \cdot pr \cdot Area_{unit})^{0.56}$ to express the dynamic process of the sediment transport capacity. Meanwhile, W is related to vegetation, soil, and land-use management (López-Vicente and Ben-Salem, 2019). The values of K , C , and P were applied considering the effect of spatial-scale variations in the underlying surface properties on sediment connectivity. Notably, W was adjusted to characterize the actual sediment quantity, which depended on the event intensity, as shown in Eq. (6).

$$W_i = 11.8 \cdot (Q_i \cdot pr \cdot Area_{unit})^{0.56} \cdot K_i \cdot C_i \cdot P_i \quad (6)$$

The functional connectivity index (FCI_k) (Eq. (7)), which reflects the relative value of connectivity associated with rainfall-runoff at the spatial and temporal scales of the watershed, was computed using the watershed outlet as the target. FCI_k was expressed using the following equation:

$$FCI_k = \log_{10} \left(\frac{11.8 \cdot (\bar{Q} \cdot \bar{pr})^{0.56} \cdot \bar{K} \cdot \bar{C} \cdot \bar{P} \cdot \sqrt{A_k}}{\sum_{i=k}^n \frac{d_i}{W_i S_i}} \right) \quad (7)$$

We normalized the FCI_k , based on Boltzmann-type sigmoid curves to facilitate comparison of the degree of sediment connectivity (DSC) in different cases (Vigiak et al., 2012). The value range was 0 to 1, calculated as follows Eq. (8):

$$DSC_{unit,k} = \left(1 + \exp \left(\frac{FCI_{0,k} - FCI_k}{k_{FCI,k}} \right) \right)^{-1} \quad (8)$$

where $FCI_{0,k}$ represents the functional connectivity metric corresponding to 0.5 in the curve. The coefficient $k_{FCI,k}$ regulates the slope of the curve; the larger the absolute value of $k_{FCI,k}$, the stronger the transition. $FCI_{0,k}$ and $k_{FCI,k}$ are calibration parameters that can be determined in light of watershed characteristics and sediment events.

3.1.3. Simulated sediment yield under rainfall event

Actually, simulated sediment yield (SY_{sim}) is a binary function related to potential soil loss ($E_{unit,k}$) and sediment connectivity ($DSC_{unit,k}$); we calculated the simulated sediment yield using the following equation:

$$SY_{sim} = \sum_k E_{unit,k} \cdot DSC_{unit,k} = \sum_k \frac{E_{unit,k}}{1 + \exp \left(\frac{FCI_{0,k} - FCI_k}{k_{FCI,k}} \right)} \quad (9)$$

To validate and calibrate the simulated sediment yield, the observed sediment yield (SY_{obs}) of a single rainfall event was calculated using the following equation (Eq. (10)):

$$SY_{obs} = \int Q_s \cdot c dt \quad (10)$$

where Q_s is the runoff at the outlet of the watershed (m^3/s) and c is the suspended sediment content (kg/m^3).

The simulated annealing algorithm (SA) was utilized to determine the optimal parameters $FCI_{0,k}$ and $k_{FCI,k}$. The objective function was set in the SA, as Eq. (11).

$$F_{max} = 1 - \frac{\sum_{j=1}^n (\log SY_{obs,j} - \log SY_{sim,j})^2}{\sum_{j=1}^n (\log SY_{obs,j} - \log SY_{ave})^2} \quad (11)$$

where SY_{ave} (ton) is the average observed sediment yield for j events.

3.2. Spatial heterogeneity analysis

To clarify whether the magnitude of erosion and the degree of sediment connectivity were consistent, we counted the areas of mismatch between erosion and sediment connectivity for the 58 rainfall events, using the geometric spacing classification. Geodetector was used to detect spatial variations in sediment yield in response to erosion and

connectivity. Specifically, the factor detector quantifies the degree of explanation calculated using the q-value (Wang et al., 2016). Geodetector was also applied to identify any interactions between erosion and sediment connectivity at spatial units, to increase or decrease the explanatory force of the sediment source, or determine whether their effects were independent.

3.3. Model evaluation and uncertainty analysis

3.3.1. Evaluation metrics

The correlation coefficient (r), Nash-Sutcliffe efficiency coefficient (NSE), and Willmott's index of agreement (WIA) were computed in accordance with the goodness-of-fit criteria, while comprehensively evaluating the prediction of the model (Krause et al., 2005; Eqs. (12–14)). Where, $\overline{SY_{obs}}$ and $\overline{SY_{sim}}$ are average observed value and average simulated value of sediment yield.

The r (Legates and McCabe (1999)) was calculated using the following equation:

$$r = \frac{\sum_{j=1}^n (SY_{sim} - \overline{SY_{sim}})(SY_{obs} - \overline{SY_{obs}})}{\sqrt{\sum_{j=1}^n (SY_{sim} - \overline{SY_{sim}})^2 \sum_{j=1}^n (SY_{obs} - \overline{SY_{obs}})^2}} \quad (12)$$

The r describes the covariance between the observed and simulated variables. The proportion of total variance explained by the model can be expressed using r^2 .

The NSE (Nash and Sutcliffe, 1970) was defined as:

$$NSE = 1 - \frac{\sum_{j=1}^n (SY_{obs} - SY_{sim})^2}{\sum_{j=1}^n (SY_{obs} - \overline{SY_{obs}})^2} \quad (13)$$

The value of NSE varies from $-\infty$ to 1, with 1 representing the perfect model performance.

The WIA (Willmott, 1981) was determined using the following equation:

$$WIA = 1 - \frac{\sum_{j=1}^n (SY_{obs} - SY_{sim})^2}{\sum_{j=1}^n (|SY_{obs} - \overline{SY_{obs}}| + |SY_{sim} - \overline{SY_{obs}}|)^2} \quad (14)$$

where WIA is a measure of the degree of error-free prediction of the model. The WIA is a standardized metric and is preferred over mean square error because it can compare different datasets more effectively. Notably, the larger the value of WIA, the higher the degree of consistency between the observed and simulated values.

3.3.2. Uncertainty analysis

The selected samples of rainfall events and the corresponding observed sediment yields could contribute to the variations in the calibration parameters when fitting the model. Therefore, we utilized leave-one-out cross-validation method to evaluate the effects of rainfall events on the forecast results (Wong, 2015). Each rainfall event sample was used as a test set, and the remaining ($n-1$) data samples were used as the corresponding training set. Therefore, when the parameters were calibrated using the SA, the results reflected the predicted fluctuations. In conclusion, the stability of the model was reflected in three aspects: range of calibration parameters ($FCI_{0,k}$, $k_{FCI,k}$), fluctuations in the NSE indexes, and relative errors (RE) of the output test results (Eqs. (13) and (15)).

$$RE = \frac{SY_{sim,j} - SY_{obs,j}}{SY_{obs,j}} \quad (15)$$

where RE reflects the deviation of the simulated sediment yield from the observed yield for the test sample.

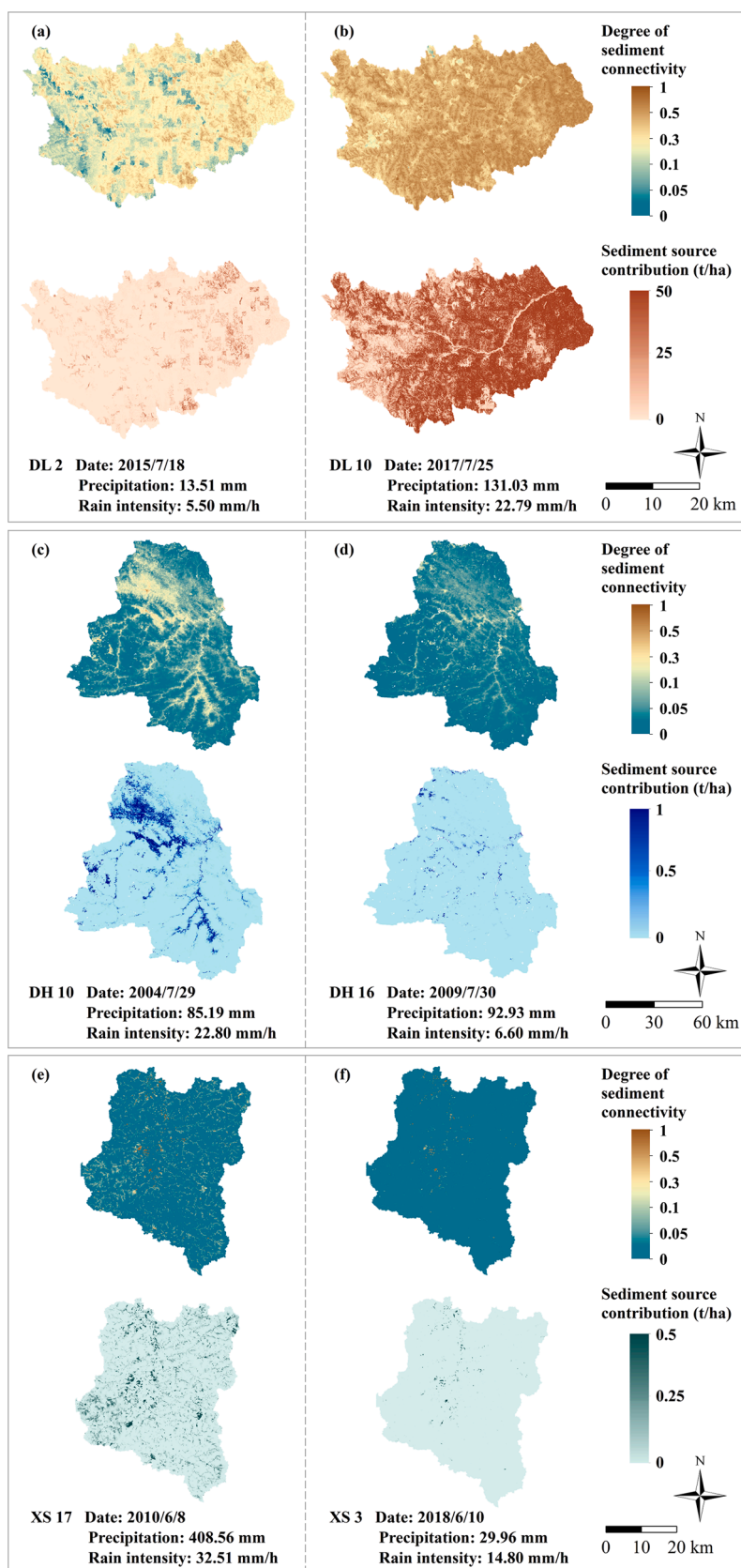


Fig. 3. Spatial variations in the sediment connectivity and sediment source contribution at unit scale for the six rainfall events in three watersheds. (a) and (b) are the 2nd and 10th events in the DL watershed, marked as DL 2, DL 10; (c) and (d) are the 10th and 16th events in the DH watershed, marked as DH 10, DH 16; (e) and (f) are the 17th and 3rd events in the XS watershed, marked as XS 17, XS 3.

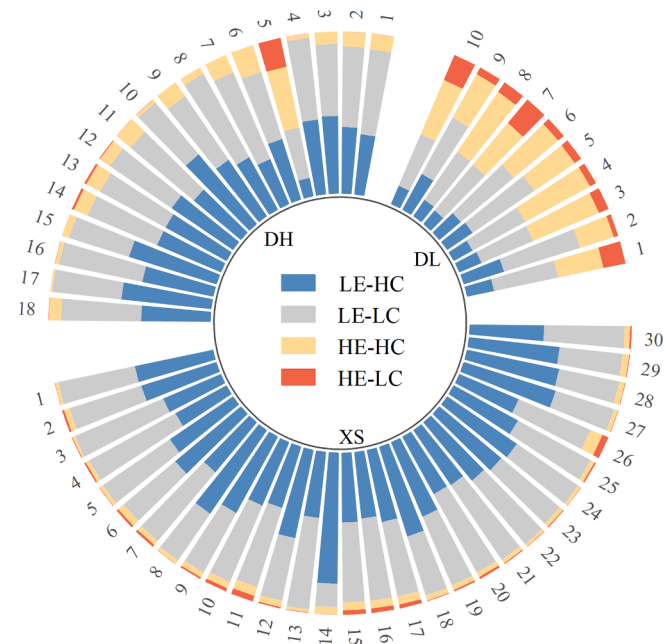


Fig. 4. Area proportions of different erosion-connection patterns. LE-HC: low erosion-high connectivity part, LE-LC: low erosion-low connectivity part, HE-HC: high erosion-high connectivity part, HE-LC: high erosion-low connectivity part.

4. Results

4.1. Spatial sediment connectivity under different rainfall regimes

The spatial variations in sediment connectivity within the watersheds were accurately identified during different rain events. We selected six events as examples for detailed comparative analysis (Fig. 3). These events occurred during the same month for each watershed to minimize other interferences. The units were connected during the low-intensity hydrological events. When the hydrological intensity increased, the units covered by the degree of sediment connectivity greater than 0.2 remained to connect and tended to expand outward (Fig. 3). In general, approximately 80 % of the area in the watershed had a sediment connectivity degree of <0.5 (Fig. 3). The spatial distribution of the sediment source was broadly consistent with the spatial pattern of the degree of sediment connectivity (Fig. 3). The low sediment contributions in most regions proved that obvious sediment deposition was exacerbated by dis-connectivity during the transport. Exceptionally, some areas having sensitive pathways kept a high contribution to the sediment yield of watershed outlets, regardless of the rainfall regime (Fig. 3(b, d and f)). In addition, the overall connectivity of the DL watershed was much higher than that of the XS watershed, even in cases where the rainfall was minimal, owing to poor vegetation coverage and the presence of several gullies, as observed in the horizontal comparison of different watersheds (Fig. 3(a and e)).

Sediment connectivity exhibited a significant nonlinear response to rainfall. Although the variation in sediment connectivity differed depending on watershed characteristics, the tenfold increase in rainfall led to significant variations in sediment sources (Fig. 3(a, b, e and f)). After extreme storm events, the sediment connectivity increased from 0.2 to 0.5, with a corresponding increase in the average sediment contribution per source area from 1.55 t/ha to 36 t/ha (Fig. 3(a and b)). In addition, the increase in rainfall improved the contribution of steep-sloped upland areas to the sediment yield at the outlet (Fig. 3(e and f)). There was also a phenomenon that the areas near the outlet and river remained relatively high connectivity, which was most obvious in the largest watershed (Fig. 3(c and d)). However, there was no high

Table 3

Effects and interaction categories of the spatial distribution of erosion and sediment connectivity degree on sediment source contribution in the three watersheds.

Events	q statistic		Interaction
	E	DSC	
DL 2	0.92	0.32	bi-enhance
DL 10	0.76	0.34	
DH 10	0.81	0.14	bi-enhance
DH 16	0.48	0.62	
XS 3	0.48	0.29	nonlinear-enhance
XS 17	0.24	0.14	

E: erosion; DSC: the degree of sediment connectivity; bi-enhance: $q(E \cap DSC) > \max(q(E), q(DSC))$; nonlinear-enhance: $q(E \cap DSC) > q(E) + q(DSC)$.

Table 4

Model optimal parameters and performance in the three different watersheds.

Watershed	$FCI_{0,k}$	$k_{FCI,k}$	r	r^2	NSE	WIA
Dali	2.33	2.99	0.98	0.96	0.74	0.89
Duhe	4.79	2.93	0.92	0.85	0.83	0.95
Xiangshui	2.02	0.96	0.85	0.72	0.70	0.92

sediment source in any riparian zone when the rain intensity was low, even though these units were highly connected. In particular, the 16th event in DH watershed (DH16) illustrated that long-duration rainfall only transported sediment to the nearby river (Fig. 3(f)).

4.2. Spatial mismatch effects between erosion and sediment connectivity

Areas with a high potential for triggering soil erosion did not portray a similar capacity for high connectivity. We identified four regional patterns: low erosion-high connectivity (LE-HC), low erosion-low connectivity (LE-LC), high erosion-high connectivity (HE-HC), and high erosion-low connectivity (HE-LC). The mismatched areas of erosion and connectivity reached an average of 43.61 %. The increase in mismatched areas significantly decreased the sediment yield ($p < 0.05$). As shown in Fig. 4, both LE-HC and LE-LC accounted for more than 50 %, with the highest proportion being 98.61 %, indicating that the sediment yield at the outlet of the watershed was mainly derived from a large number of units having low erosion rates. The HE-LC pattern appeared the least in the watersheds. Moreover, an increase in the area of the HE-HC regions contributed to a significant increase in sediment yield. The DL watershed, which represented an erosion-prone area of the Loess Plateau, had a higher share of HE-HC than the DH watershed, and the lowest percentage of the high-erosion area was observed in the XS watershed (Fig. 4).

Erosion and connectivity were strongly correlated with the distribution of sediment sources. The q values of erosion and sediment connectivity were statistically significant, and both were positively correlated with the sediment yield at spatial units ($p < 0.05$). Erosion had a higher control on the spatial distribution of sediment source contributions compared to sediment connectivity, except for DH16 (Table 3). Further, the spatial distribution of erosion was less influential in controlling the sediment yield when the scale of erosion decreased (Table 3). The interaction between erosion and sediment connectivity was bi-enhanced in the DL and DH watersheds, suggesting that the combined effect of the two factors was greater than that of each factor individually. In addition, the interaction effect of erosion and sediment connectivity was a nonlinear enhancement greater than the sum of both the factors in the XS watershed. This was consistent with previous results that the HE-HL pattern increased sediment yield significantly.

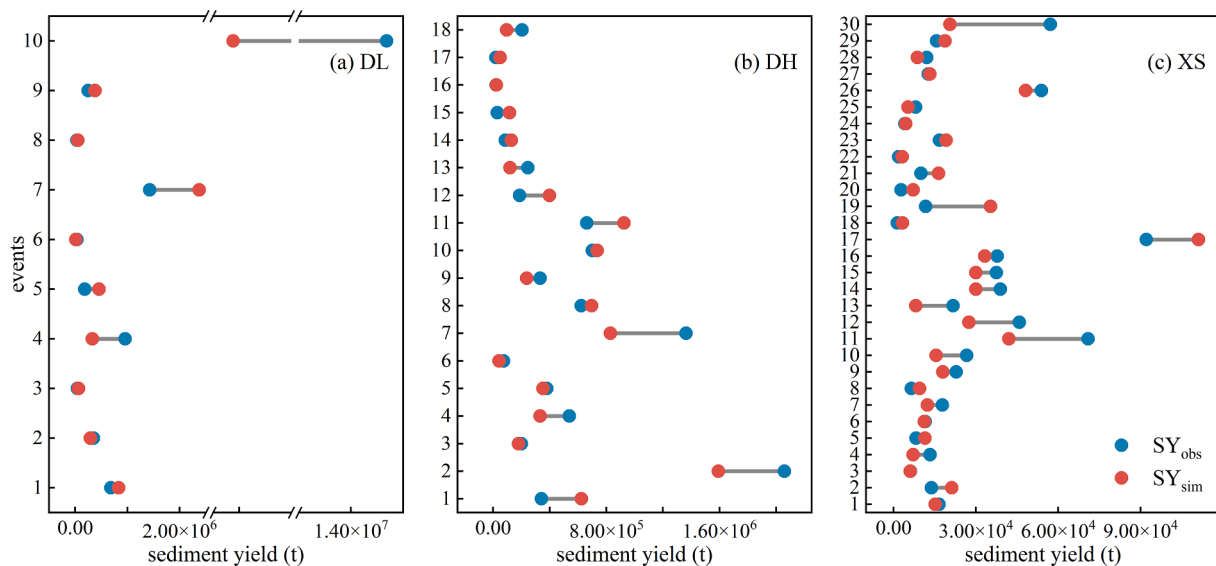


Fig. 5. Simulated and observed values of sediment yield for different events.

4.3. Evaluation of method performance

The results of sediment yield, obtained using 58 storm events in the three watersheds, were considered in detail to identify the model performance. As shown in Table 4, the values of each evaluation metric indicated a good model performance from different perspectives. The r -values for the DL, DH, and XS watersheds were 0.98, 0.92, and 0.85, respectively, suggesting that the prediction model fitted the observed data well. The $r^2 > 0.72$ indicated that the prediction error of the model was low. In addition, the NSE was greater than 0.7, indicating that the simulated sediment yields were reliable. The WIA values, which serve as a measure of the degree of freedom from error, were also above 0.89.

Quantitative comparisons of the observed versus simulated values provided credibility and pointed out possible miscalculations. The simulation error for the 10th event in DL watershed (DL10) was large (Fig. 5(a)), with the observed sediment yield being 1.47×10^7 tons and the simulation result being 7.88×10^6 tons. This may be because the results for DL10 represented the amount of sediment yield caused by an extreme rainstorm once over a hundred years. When the model was applied to the DH watershed, the maximum difference between the simulated sediment yield and observed value was 5.34×10^5 tons, with a relative error of 0.39. The best simulated sediment yield was that for DH16, with an error of 1.01×10^3 tons and a relative error of 0.04 (Fig. 5(b)). Moreover, the model was well adapted to 30 events in the XS watershed and was capable of predicting the trends in sediment yield under different rainfall conditions (Fig. 5(c)). Overall, the model results were within the acceptable ranges.

The effect of rainfall on the simulation results in the validation set was reflected. The parameter $FCI_{0,k}$ varied significantly in the DL watershed, with the coefficient of variation (CV) being 0.14 (Fig. 6(a)). The CV values of the parameter $FCI_{0,k}$ in the DL and XS watersheds were 0.02 and 0.05, respectively. In addition, the 6th event had the greatest effect on the value of $k_{FCI,k}$ in the DL watershed (red circle in Fig. 6(b)). The difference in rainfall events had the greatest effect on the $k_{FCI,k}$ values in the XS watershed (Fig. 6(c)). Parameter $FCI_{0,k}$ was more sensitive than $k_{FCI,k}$, resulting in larger fluctuations in the simulated sediment yield (Fig. 6). Although the simulation of the sediment yield for the special single event was not accurate, the overall fit of the model was stable (NSE > 0.78) (Fig. 6).

5. Discussion

5.1. Control element analysis for sediment connectivity

At the event scale, the underlying surface factors that are invariant in the short term, such as geomorphology and soil, and vegetation cover varying over months, are regarded as stationary factors that affect sediment connectivity. The differences in the degree of sediment connectivity between the three watersheds highlighted the long-term influence of watershed structure on sediment transport (Fig. 3). Topographic configurations and geomorphic sequences (i.e., channel slope, confinement, and stepped channels) control the longitudinal and lateral transport of sediments (Scorpio et al., 2022). In general, there was dense forest cover on the steep slopes in the watersheds, which contributed to reducing erosion capacity and sediment connectivity (Mishra et al., 2019). This explained the much lower sediment yield of the XS watershed, compared to that of the DL watershed. Similarly, sediment connectivity was below 0.1 in the southern portion of the DH watershed, despite the steep topography (Kalantari et al., 2019). Vegetation provided greater benefits than runoff control, in terms of sediment reduction during storms. Additionally, sediment connectivity in the source area involves not only coupling between hillslopes and target outlets, but also between tributaries, river beaches, and channels (Czuba et al., 2017). Therefore, the sediment source contribution was always high at sensitive sites near the river and target outlet (Fig. 3). In contrast, a large amount of sediment released from the hillside was redeposited in the downslope or buffer zone of the floodplain.

Rainfall regimes are dynamic factors that affect sediment connectivity. The rainfall-driven hydrological processes ultimately determine the frequency, distribution, and magnitude of actual sediment transport in the watershed. The spatial distribution and regime variations in rainfall complicate sediment connectivity and make it difficult to track (Wainwright et al., 2011). Our study revealed that the total amount of rainfall, rain intensity, and rainfall duration controlled the degree of sediment connectivity from different aspects, which caused the sediment supply hotspot areas to display various patterns in different rainfall regimes (Fig. 3). The total rainfall and rainfall duration were critical for the cascading effect from the steep uplands in the watershed. Only large-scale events could activate slope-channel-outlet sediment connectivity throughout the watershed (Fryirs, 2013). Another study indicated that the center location of the storm, relative to the watershed outlet, affected sediment transport activities (Sigaroodi and Chen, 2016). A high rainfall intensity would control more areas that had high

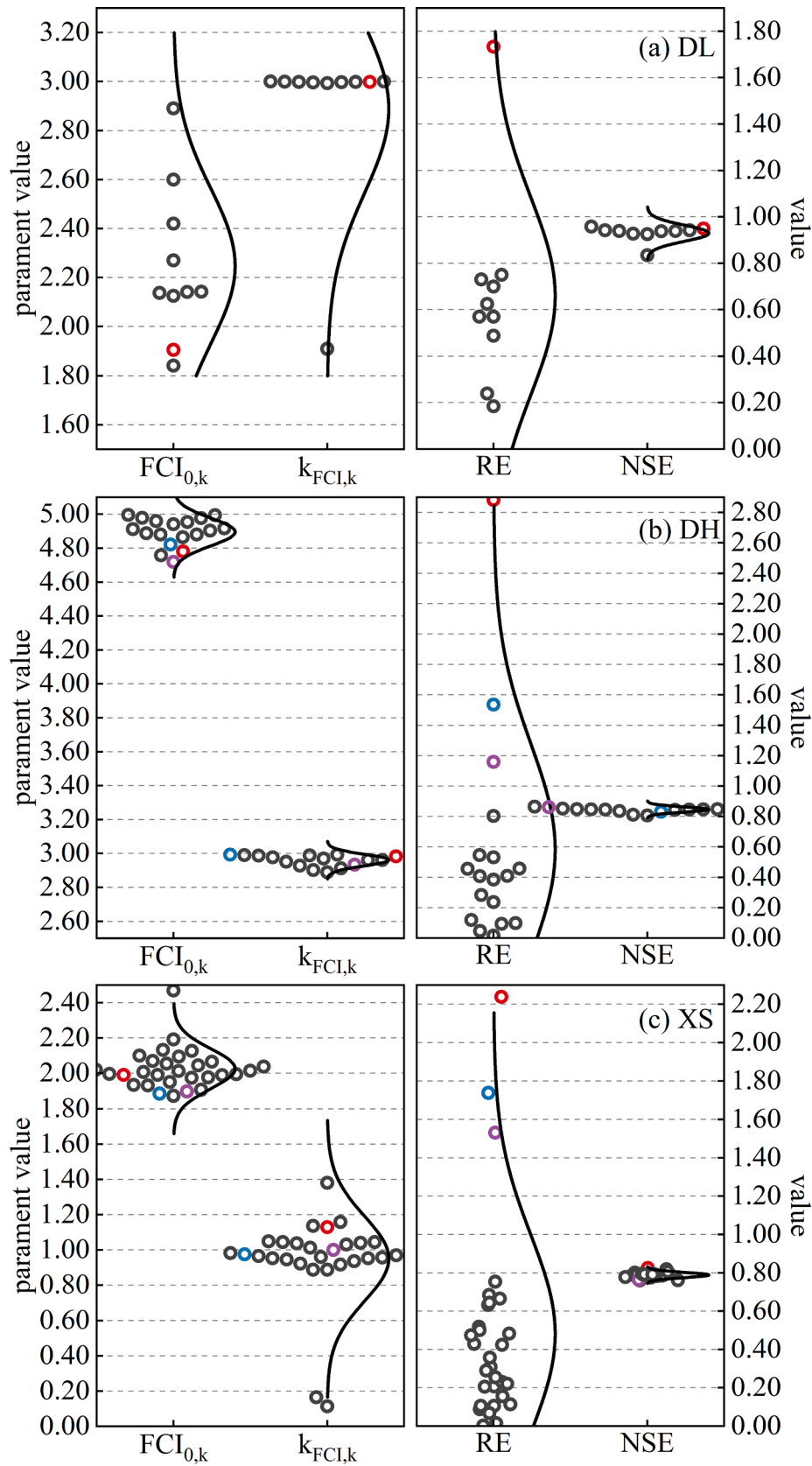


Fig. 6. Range distribution of the parameters $FCI_{0,k}$ and $k_{FCI,k}$, the relative error (RE) in the validation data, and the fluctuations in Nash-Sutcliffe efficiency coefficient (NSE). The red circle indicates the value corresponding to the maximum relative error of the validation data, followed by the blue and purple circles. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

connectivity, resulting in a shorter distance and rapid transport of the temporarily stored river sediment, as illustrated in Fig. 3(c and d). Furthermore, the 2nd event (13,741 tons of sediment yield) and 3rd event (6,170 tons of sediment yield) occurred consecutively on June 5 and June 10, 2018 in the XS watershed (Fig. 5(c)). Erosive events export large amounts of sediments that are temporarily stored in the channel during the initial stages of flooding (Chartin et al., 2017). Part of the sediment was only transported to the slope bottom in the 2nd event that received 118 mm of rainfall and had a rain intensity of 18 mm/h. Then, the 3rd event (30 mm rainfall and 14 mm/h rain intensity) remobilized the previously deposited sediment, thereby causing high sediment discharge at the outlet under this small hydrological condition. An investigation of two floods in Italian mountain catchments similarly found that, despite the massive activation of sediment sources on hill-sides in 2017, sediment storages on talus slopes were still available to provide sediment in 2020 (Scorpio et al., 2022). Thus, our results support the evidence from previous studies (Mahoney et al., 2021), reiterating that antecedent hydrologic events can be used as stencils for larger events. Previous studies also demonstrated that a combination of rainfall and runoff was capable of predicting the total suspended sediment load (Tuset et al., 2016). In particular, it was important to consider the proportion of rainfall converted into runoff by comparing the DL and XS watersheds. For example, a tenfold increase in total rainfall resulted in a nonlinear increase in rainfall erosivity and connectivity, leading to a 42-fold increase in sediment yield (Fig. 5(a)). Therefore, various rainfall events may increase or decrease the flow energy available for sediment separation and transport at different units, thus, triggering uncertainty in sediment yield when calculating the parameters (Fig. 6).

5.2. Necessity of sediment connectivity in transport assessment

The dynamic assessment of sediment connectivity is an essential step toward achieving a comprehensive understanding of the transport between the sediment source and sink areas. A low sediment yield at the watershed scale may not mean that high erosion rates do not exist in the watershed; instead, the reason can be a mismatch between erosion and connectivity (Fig. 4). Erosion represents the amount of sediment available, whereas connectivity describes the accessibility of the sediment (Najafi et al., 2021). Erosion serves as a prerequisite to sediment supply, which is why the sediment source distribution is more dependent on erosion (Table 3). Moreover, the spatial simulation of sediment connectivity for rainfall events clarified the strength of the sediment transport capacity, which determined the rate of sediment that reached the watershed outlet (Fig. 3). We verified the limiting intensity of restrictions on erosion and connectivity, indicating that the comprehensive influence of these two factors enhanced the explanation of the spatial distribution of the sediment sources (Table 3). Therefore, the erosion source distribution and the degree of sediment connectivity require to be coupled when calculating sediment yield at the watershed scale. Some studies also integrated connectivity in erosion models and confirmed their validity (Rajbanshi and Bhattacharya, 2020; Woznicki et al., 2020). For instance, the sediment yield in the Yanhe River catchment was estimated by combining IC with the RUSLE model, thereby assessing the impact of soil conservation measures (Zhao et al., 2020). However, IC represents only the potential sediment transport rate of the landscape, and RUSLE is restricted to the annual scale; thus, these models have low sensitivity to short-term rainfall variations (Michalek et al., 2021). Moreover, the parameters in these studies used the fixed values proposed by Vigiak et al. (2012). Our results showed that different watersheds and rainfall events could cause the parameters to fluctuate (Fig. 6). Once the watershed landscape features and rainfall regimes varied, the parameters should be changed; otherwise, the transport processes may be distorted at partial positions. In this study, dynamic indicators were added to modify the weighting factor (W), and the parameters were calibrated to replace the empirical values, based on the watershed characteristics and known events. The incorporation of

the dynamic assessment of sediment connectivity and the MUSLE model bridges the gap between simulated and actual sediment transport under different rainfall events.

By coupling sediment connectivity with the classic erosion formula, we provided a straightforward resolution to quantitatively identify the critical areas of sediment load, especially for data-sparse areas. The simulated sediment yield, estimated by sediment connectivity and in-situ erosion, portrayed excellent and stable performance (Table 4, Fig. 6). The coupled connectivity and MUSLE were convenient to obtain and calibrate the parameters, reducing the impact of physical structure on simulation accuracy, compared with distributed process-based models, such as the Watershed Erosion Prediction Project (WEPP) and Limburg Soil Erosion Model (LISEM) (de Barros et al., 2014; Shen et al., 2009). In addition, although the Soil and Water Assessment Tool (SWAT) employing the MUSLE considered the variations in the SDR caused by surface runoff over time. The average SDR value was still used in space, which could hardly reflect the spatial distribution of sediment transport (Shi and Huang, 2021). Our method implemented the mapping of specific pathways from erosion sites to watershed outlets and identified different sediment sources for each rainfall event. It is possible to forecast soil loss risks using rainfall data in watersheds without long-term hydrological sediment monitoring.

6. Conclusions

The quantification of sediment connectivity assisted in mapping the spatial distribution of sediment contribution per unit of watershed at the rainfall event scale. We confirmed that the total amount and duration of rainfall were crucial for the cascading effect of long-distance sediment transport in the watershed. Whereas rain intensity controlled the short-distance and rapid migration of sediments that were temporarily stored in the riverbanks. The proximal riverbank was a hotspot source affecting sediment yield at the watershed outlet. Erosion and connectivity reflect the sediment loss issues in terms of both sediment denudation and downstream transport loss, respectively. Mismatched areas with respect to erosion and sediment connectivity were identified in all three watersheds. Notably, high-erosion areas did not contribute to the major sediment yield sites when they did not have high connectivity. The detection of erosion and connectivity interaction effects confirmed that considering both factors improved the understanding of the spatial distribution in sediment source contribution. The simulated sediment yields of rainfall events that combined erosion and connectivity were in good agreement with the observed values. Therefore, our study provides sufficient quantitative information on sediment connectivity in different rainfall regimes to identify sediment hotspots, which may enable managers to adjust ecological measures and reduce the risk of sediment transport in the watershed. Coupling erosion and sediment connectivity for the purpose of source control and process optimization can be recommended in watershed sediment management. Future work to investigate the location of sediment deposition and transport duration during rainfall occurrence, as well as field sediment tracing involving validation, is needed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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